

● EASY

● INFORMATION AND IDEAS

Information and Ideas

08

10 questions · ~15 min

Q1

INFORMATION AND IDEAS

Many scientists have believed that giraffes are solitary creatures, preferring to spend their time alone instead of with others. But observations of giraffes and their behavior in recent years has suggested that these animals may be more social than we once thought. For example, scientists Zoe Muller and Stephen Harris claim that giraffes may even help each other care for one another's newborns.

Which finding, if true, would most directly support Muller and Harris's conclusion?

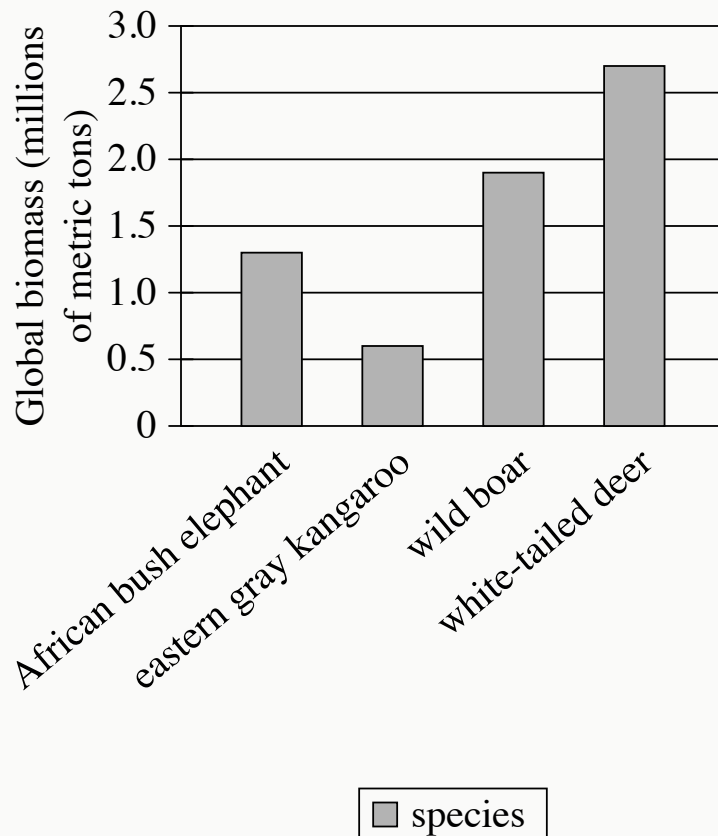
- A. Female giraffes have been observed feeding young giraffes that aren't their direct offspring.
- B. Confrontations between a younger and an older male giraffe are frequently observed.
- C. Some female giraffes have been observed sniffing and licking their newborn offspring.
- D. Giraffes are able to make sounds but are rarely observed communicating with others.

Xin Wang and colleagues have discovered the earliest known example of a flower bud in a 164-million-year-old plant fossil in China. The researchers have named the new species *Florigerminis jurassica*. They believe that the discovery pushes the emergence of flowering plants, or angiosperms, back to the Jurassic period, which occurred between 145 million and 201 million years ago.

According to the text, how old was the fossil that Wang and colleagues discovered?

- A. 150 million years old
- B. 145 million years old
- C. 164 million years old
- D. 201 million years old

Top Four Species of Wild Land Mammals by Global Biomass



- The Global biomass data for the 4 bars are as follows:
 - African bush elephant: 1.3 million metric tons
 - eastern gray kangaroo: 0.6 million metric tons
 - wild boar: 1.9 million metric tons
 - white-tailed deer: 2.7 million metric tons

Global biomass is the total mass of living material, such as animals and plants, on Earth. A team of scientists estimated the global biomass, by species, of various wild land mammals. The team found that the species with the highest global biomass is the _____ blank

Which choice most effectively uses data from the graph to complete the sentence?

- A. wild boar.
- B. eastern gray kangaroo.
- C. African bush elephant.
- D. white-tailed deer.

The following text is from Beatrice Harraden's 1894 novel *Ships that Pass in the Night*.

In an old second-hand bookshop in London, an old man sat reading Gibbon's *History of Rome*. He did not put down his book when the postman brought him a letter. He just glanced indifferently at the letter, and impatiently at the postman. Zerviah Holme did not like to be interrupted when he was reading Gibbon; and as he was always reading Gibbon, an interruption was always regarded by him as an insult.

Based on the text, how did Zerviah Holme most likely feel when the letter was delivered?

- A. He felt relieved because he had been expecting an important letter.
- B. He felt excited because the letter was from a good friend.
- C. He felt sad because the postman did not stop to talk with him before leaving.
- D. He felt annoyed because he was interrupted while reading his favorite author.

Biologists have generally believed that the diet of jaguars consists mostly of land-based mammals, but researchers studying a population of jaguars living in the Brazilian Pantanal, a tropical wetland, claim that jaguars can survive on a diet of more fish and aquatic reptiles than mammals.

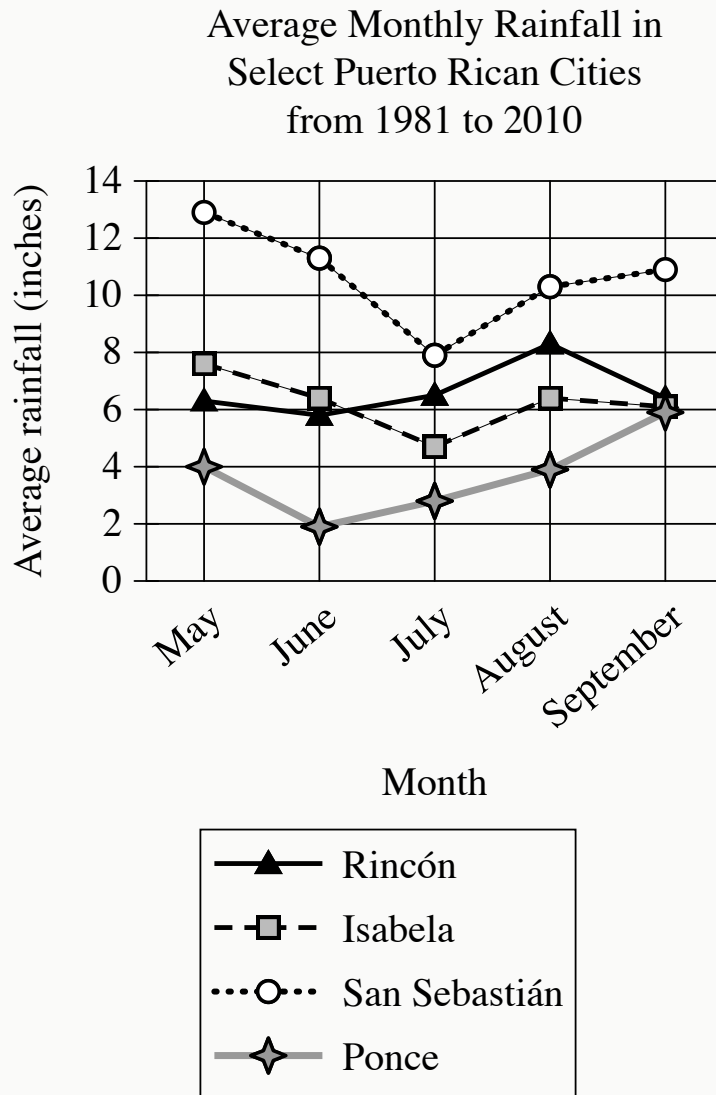
Which finding, if true, would most directly support the researchers' claim?

- A. Aquatic reptile and fish remains were detected more often than were mammal remains in an analysis of jaguar waste matter found in the area.
- B. Exceptionally high numbers of yacare caiman, an aquatic reptile, were found living in the area.
- C. Aquatic prey like the reptiles and fish in the area provide a source of omega-3 fatty acids that aren't often found in land mammals.
- D. When jaguars in the area preyed on mammals, they tended to prefer semiaquatic ones like capybaras.

Oluwaseyi Moejoh cofounded U-recycle Initiative Africa when she was only a teenager. Moejoh and her team founded the organization to teach young people how their actions affect the environment and why recycling is important. For example, the organization put on an exhibit of art made using recycled materials.

According to the text, what is one reason Moejoh and others founded U-recycle Initiative Africa?

- A. To bring attention to overlooked African artists
- B. To teach young people why recycling is important
- C. To help adults gain important outdoor skills
- D. To give teenagers advice about starting businesses



- The following 4 lines are shown:
 - Rincón
 - Isabela
 - San Sebastián
 - Ponce
- The Rincón line:
 - Begins at May, 6.3 inches
 - Falls gradually to June, 5.8 inches

- Rises gradually to July, 6.5 inches
- Rises sharply to August, 8.3 inches
- Falls sharply to September, 6.4 inches
- The Isabela line:
 - Begins at May, 7.6 inches
 - Falls gradually to June, 6.4 inches
 - Falls gradually to July, 4.7 inches
 - Rises gradually to August, 6.4 inches
 - Falls gradually to September, 6.1 inches
- The San Sebastián line:
 - Begins at May, 12.9 inches
 - Falls gradually to June, 11.3 inches
 - Falls sharply to July, 7.9 inches
 - Rises sharply to August, 10.3 inches
 - Rises gradually to September, 10.9 inches
- The Ponce line:
 - Begins at May, 4 inches
 - Falls sharply to June, 1.9 inches
 - Rises gradually to July, 2.8 inches
 - Rises gradually to August, 3.9 inches
 - Rises gradually to September, 5.9 inches

A student is presenting average monthly rainfall totals in various Puerto Rican cities for a science class. During the presentation, the student notes that in September _____ blank

Which choice most effectively uses data from the graph to complete the statement?

- A. Rincón’s average rainfall is greater than that of Isabela, San Sebastián, and Ponce.
- B. Rincón and Ponce have an average rainfall of about 5 inches, and Isabela and San Sebastián have an average rainfall of about 10 inches.
- C. Rincón, Ponce, and Isabela each have an average rainfall below 8 inches, but San Sebastián’s average rainfall that month is greater than 8 inches.

- D. Rincón has a similar average rainfall to Isabela, and Ponce has a similar average rainfall to San Sebastián.

The following text is adapted from Sylvia Acevedo's 2018 memoir *Path to the Stars: My Journey from Girl Scout to Rocket Scientist*. The narrator is traveling by car with her family to Mexico City. Mario and Laura are her brother and sister.

Mario and I played games to see how many different license plates we could spot, and Laura liked to look for children in the back seats of the cars we passed. We were used to the forty-five-minute drive to El Paso and familiar with the six-hour ride to Chihuahua, but I wondered what the long journey to Mexico City would be like.

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According to the text, what did the narrator and Mario do while riding in the car?

- A. They read books.
- B. They sang songs.
- C. They went to sleep.
- D. They played games.

Scientists have long believed that giraffes are mostly silent and communicate only visually with one another. But biologist Angela Stöger and her team analyzed hundreds of hours of recordings of giraffes in three European zoos and found that giraffes make a very low-pitched humming sound. The researchers claim that the giraffes use these sounds to communicate when it's not possible for them to signal one another visually.

Which finding, if true, would most directly support Stöger and her team's claim?

- A. Giraffes have an excellent sense of vision and can see in color.
- B. The giraffes only produced the humming sounds at night when they couldn't see one another.
- C. Wild giraffes have never been recorded making humming sounds.
- D. Researchers observed other animals in European zoos humming.

The average age at which people in the United States start businesses is 35. Economist Andrés Hincapié studied why young adults are relatively less likely to start businesses and whether there are ways to increase entrepreneurship in early adulthood. Hincapié found that one impediment is lack of knowledge about the practical details of how businesses are started; he further found that simply providing young adults with good informational resources on the topic significantly alleviates this problem.

Based on the text, what would Hincapié most likely say is a promising way to increase entrepreneurship in early adulthood?

- A. Creating social networks of young adults who are interested in starting a business
- B. Encouraging young adults to brainstorm business ideas
- C. Providing young adults with practical information about how to start a business
- D. Giving young adults training opportunities at a variety of businesses